DATA STRUCTURES, FEATURE VECTORS, AND EXPORT/SHARING FORMATS

Purpose of Data Structure Disclosure

The longitudinal penile physiology monitoring system may generate structured data representations enabling efficient storage, transmission, interpretation, and analysis of physiologic information. The disclosure of data structures herein supports computational implementation flexibility while preserving broad conceptual scope.

Physiologic Feature Vectors

Sensor measurements obtained from Node 1 and companion nodes may be converted into feature vectors representing physiologic characteristics over time. Feature vectors may include mechanical deformation parameters, temporal response features, thermal measurements, motion context indicators, perfusion-derived values, and derived indices generated through sensor fusion.

Feature vectors may represent instantaneous physiologic state, aggregated session summaries, or longitudinal behavioral patterns derived from extended monitoring.

Hierarchical Data Models

Data may be organized using hierarchical structures including raw sensor layers, processed signal layers, derived metric layers, and abstract profile layers. Hierarchical organization may permit selective sharing of higher-level insights without exposing raw biometric signals.

Temporal Data Encoding

Physiologic datasets may incorporate timestamp encoding, sequence identifiers, event markers, or synchronization metadata enabling reconstruction of physiologic timelines across multiple devices or monitoring sessions.

Export and Interoperability Formats

Data may be exported using standardized or proprietary formats including encrypted datasets, compressed feature representations, anonymized summaries, or application programming interface (API) compatible structures enabling interoperability with health platforms, research systems, or external computational environments.

Privacy-Preserving Data Sharing

Exported data may exclude identifiable anatomical information and instead transmit abstracted metrics, symbolic profiles, compatibility indicators, or derived performance scores. Permission controls may govern access to varying data layers.

Future Computational Compatibility

The system may support evolving computational paradigms including machine learning pipelines, distributed cloud processing, edge inference architectures, or future artificial intelligence frameworks capable of utilizing structured physiologic datasets.

The data structure embodiments described herein are illustrative and non-limiting and encompass present and future methods of representing wearable-derived physiologic information in structured digital formats.